

What each slide claims, and why

One page per slide. What it asserts, the evidence behind it, and the sentence you should be able to say if a supervisor stops you in a corridor.

SLIDES	DATA BEHIND THEM	PHASE	READ TIME
Five	168 min, 8 drives	C, step 3	Ten minutes

BEFORE YOU READ THE SLIDES

The whole project in three sentences

Engines contain slow things that heat damages. A controller that only sees *now* has to wait for the temperature to climb before backing off, by which time the damage is done — but one that knows a hill is coming can protect before the heat arrives.

How much that anticipation is worth depends on one ratio: how far ahead you can see, divided by how slowly the protected thing responds. That ratio is H/τ , and nobody has mapped it. That gap is our project.

We are not proposing a better engine controller. That is commercially solved — Mercedes ships predictive powertrain control on trucks today. We propose a *criterion* for when preview information is worth acquiring at all. If someone asks “isn’t this already done?”, that distinction is the whole answer.

SLIDE 1 Problem definition

the gap

WHAT THE SLIDE CLAIMS

Every control engineer faces the same decision before building anything: is it worth paying for information about the future? Sensors, maps and GPS lookahead all cost money, weight and complexity. There is no principled way to answer that question in advance, so it gets answered by guesswork.

WHY IT IS A REAL GAP AND NOT A MADE-UP ONE

Preview control theory proves the benefit of preview *converges* as the horizon grows. It does not say how fast, or how much you get at a particular horizon. So a designer knows more preview is never worse, which is not a useful thing to know when deciding whether to fit a sensor.

SAY THIS

“We are not asking how to use preview. We are asking when it is worth having — and that question has a dimensionless answer nobody has measured.”

WHAT THE SLIDE CLAIMS

Measure the value of preview across two physically unrelated systems, sweep the ratio it should depend on, and see whether the measurements collapse onto a single curve.

PLANT 1	A validated engine simulator — 8 of 11 published quantities inside band, 2.8 % load error against our own car.
AGENT	A constrained reinforcement-learning agent, measured against three baselines: the stock ECU, a reactive rule, and the same agent with preview switched off .
PLANT 2	An EV battery pack, driven by identical agent code, at an H/τ two orders of magnitude away.
SWEEP	Vary the horizon and the time constant; plot preview value against their ratio.

THE BAR CHART ON THAT SLIDE

Holding the preview horizon fixed at 30 seconds and changing *only* the turbine's thermal mass moved preview value from **16.5 to 26.0 points**. Same policies, same actuators, one ratio changed.

SAY THIS

"The claim is falsifiable. Points from both plants should land on one curve. If they do not, the criterion is wrong or incomplete — and reporting that is still a result."

WHAT THE SLIDE CLAIMS

The simulator is not a textbook exercise. It is checked against 168 minutes of read-only OBD-II logging from our own 2023 GR Supra, across eight recording sessions.

168 MIN	eight drives, 3.5–6.6 Hz sampling
25 CH	channels recorded, all confirmed carrying live data
22 PTS	distinct steady operating points, 1551–5053 rpm
2.8 %	load error, model against measurement

THE ONE THING TO UNDERSTAND ABOUT THE CHART

The two lines overlay because the model reproduces the load trend after a *single* calibration constant — and that constant is BMW's own load normalisation, not a model parameter we tuned. We fitted one number, not twenty-two.

THE FINDING WORTH TELLING PEOPLE ABOUT

Sixty seconds of parked idle proved the channel labelled “intake manifold absolute pressure” is actually a **pre-throttle sensor**. At warm idle it reads near ambient, where a throttled engine must sit near a third of ambient. Feeding it to the model gives **75 % air-mass error**. Inverting the air-mass channel instead gives 2.8 %.

SAY THIS

“Validated 31–82 kPa, stoichiometric, naturally aspirated. The boosted region is *not* validated against the car — steady points need steady driving, and steady driving is light-load driving. That is stated in the thesis, not hidden.”

SLIDE 4 Paper review

the positioning

WHAT THE SLIDE CLAIMS

Three pieces of prior work sit closest to ours. Each establishes something real, and each leaves the same thing open.

THEORY	Liu & Seiler. Regret bounds for H_∞ preview controllers — proves preview benefit converges as the horizon grows. Linear time-invariant only; no decay rate, no magnitude at a given horizon.
PRODUCT	Daimler PPC. Map-based lookahead controlling truck cruise and gear selection. Proves preview control is commercially viable; it is engineering, not a criterion.
RL	Bedei et al. The closest hardware-validated constrained-RL engine work — and it explicitly has <i>no</i> preview.

WHY NAMING THE NEAREST PAPER HELPS US

An examiner who knows preview control will think of Liu and Seiler within a minute. Naming it first, and saying precisely what it leaves open, is worth far more than claiming no prior work exists.

SAY THIS

“Theory proves convergence but gives no magnitude. Industry ships it without measuring when it pays. The RL work has no preview. The gap is the measurement, and that is what we are doing.”

SLIDE 5 Deliverables and challenges

the honesty

FOUR DELIVERABLES

A validated engine simulator; a constrained RL agent with three baselines; a second plant driven by unmodified agent code; and the H/τ figure with both plants on one set of axes.

THREE CHALLENGES — AND WE RAISE ALL THREE OURSELVES

1. **The data cannot reach the boosted region.** Validated against published correlations up there, and the thesis says so.
2. **We refuted one of our own hypotheses.** We predicted preview value would grow with cost-function convexity. It fell — 36.8 points down to 26.0. It goes in the results chapter exactly as it came out.

3. **We caught our own largest error.** The simulator defaulted to a 2.0 L four-cylinder for three weeks while every document said 3.0 L six. Corrected, and every affected number regenerated.

SAY THIS, OUT LOUD, BEFORE ANYONE FINDS IT

“A student who reports a result that went against them is trusted on everything else they claim.” That last point does not weaken the project — it sharpens the question it was always asking: which systems make preview worth having?

Backup

The four questions you will actually be asked

Q1 How do you know the simulator is right?

Two independent checks. Against published combustion figures: 8 of 11 quantities inside band, and we report the three misses with reasons rather than tuning to close them. Against the car: 2.8 % load error over 22 operating points from 168 minutes of logging.

Q2 Isn't predictive engine control already done?

Yes, and we say so on slide 4. We are not proposing a controller. We are measuring when preview is worth acquiring — the question that comes *before* building one.

Q3 How do you know the advantage comes from preview and not from a better agent?

This is the strongest answer we have. We run the identical agent with the preview features zeroed. In the hand-written version, the blinded policy lands on the reactive policy's score *to the decimal* — 548.6 against 548.6. That identity has held across two engines, two scenarios and two protection thresholds.

Q4 What would have falsified your claim?

If preview value did not collapse onto one curve against H/τ when plotted from two unrelated plants. We have already refuted one of our own sub-hypotheses and kept the result, so the answer is not hypothetical.

Every number in the deck is printed by a script. Run `python verify_docs.py` in the repository; it recomputes twenty-two published figures from the shipped data and fails if any has drifted. If a slide disagrees with that script, the script is right and the slide needs rebuilding.